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Blood Group Identification by Analyzing Fingerprints using ResNet-50

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***Abstract*—In the domain of medical diagnostics, identification of blood groups accurately and efficiently is essential for various clinical procedures, including transfusions and organ transplants. Traditional invasive procedures involve taking a blood sample from individuals, which can be uncomfortable and may cause infections. Additionally, testing blood samples in the laboratory is time-consuming. The present approach proposes a non-invasive method for blood group identification using deep learning and fingerprint images. Fingerprints are unique patterns found on our fingertips that help in identifying individuals. By using these unique and unchangeable features of fingerprints, every individual's blood group can be identified. The proposed approach leverages deep learning model ResNet-50, which is well known for its significance in image classification, to predict blood groups efficiently. The model achieved an accuracy of 88%, confirming its effectiveness in blood group identification. This approach addresses a critical need in the medical community, providing a seamless and technologically advanced solution for blood group identification**

Keywords: Fingerprints, minutiae patterns, medical diagnostics, transfusions, ResNet-50.

I. Introduction

Accurate and efficient identification of blood types is crucial, since the clinical consequences of this knowledge are significant, ranging from blood transfusions to organ transplants to emergency medical situations. Traditional blood typing usually involves an invasive procedure, a needle, and a blood sample from the patient, which is uncomfortable, possibly infectious, and a waste of time and money for lab testing and analysis . In emergencies when immediate recognition is required, these waits and discomforts can be truly difficult. Due to these issues, there is increasing demand for non-invasive methods [1-2] to accurately and quickly type blood without causing pain to the patient or delaying procedures.

The approach is a novel non-invasive approach that involves the use of the ResNet-50 deep learning model to process fingerprint images and predict blood types [3] among the eight ABO types (A+, A-, B+, B-, AB+, AB-, O+ and O- . Each person has unique fingerprints with intricate patterns like arches, loops, and whorls,[4] therefore they make a great form of biometric information [5-7]. There have been studies that show a correlation [8] between these fingerprint patterns and certain blood types. This method uses these natural fingerprint characteristics to try and determine blood types from fingerprint images [9].

ResNet-50 is a convolutional neural network renowned for its impressive performance in image classification tasks, and it is utilized here to analyze fingerprint images. This architecture happens to be especially good at picking up on the little nuances and intricacies involved in fingerprint patterns, so it should serve well for this purpose. Using this deep learning method, the new system can determine a person's blood type without actually having to draw blood. This non-invasive method is a great stride in diagnostic technology, where the safety and comfort of the patient is a priority, along with the short amount of time it takes to reach a diagnosis.

This method, which is fast, accurate, and non-invasive, could greatly improve patient care in a wide range of clinical settings–from the doctor's office to the emergency room–by replacing the conventional method of blood group testing. It is an interesting blend of fingerprint biometrics and advanced image processing, the medical field is going into a new direction, a direction of convenience and technological sophistication, but yet it still fills a vital medical need.

A. Objectives of the study:

- To collect a diverse dataset of fingerprint images annotated with blood group information.
- To preprocess the fingerprint images to enhance clarity and remove noise, ensuring optimal input for the ResNet-50 architecture.
- To implement the ResNet-50 architecture for feature extraction from fingerprint images, leveraging its deep learning capabilities to capture intricate patterns indicative of blood group information.
- To train and fine tune the ResNet-50 model for best accuracy and efficiency.

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- To evaluate model performance using standard metrics and cross-validation techniques.

B. Scope of the study:

- Limited to blood group identification from fingerprints without any beyond tangential research such as iris recognition or DNA analysis.
- This approach focuses on identifying the ABO blood groups, including A+, A-, B+, B-, AB+, AB-, O+, and O-, through the analysis of fingerprints.

II. LITERATURE REVIEW

A. Blood Group Determination Using Fingerprint

Nihar et al. [10] presented a new method of blood typing through finger print examination. This study is about the relationship between finger ridge patterns and blood types and how the use of biometrics can be used in the medical field. It works by identifying certain antigens associated with blood types from a fingerprint, and therefore it would be a non-invasive way of doing a blood test. The pre-trained architectures of AlexNet and LeNet-5 are used for classifying the input fingerprint data.

B. Fingerprint Based Blood Group Prediction Using Deep Learning

The study by Swathi et al. [11] introduces a new method of predicting blood types from fingerprints and machine learning. The research uses Convolutional Neural Networks (CNNs) to pull out intricate features from fingerprint pictures to predict blood type. It also uses Multiple Linear Regression with OLS (Ordinary Least Squares) and has a 62% success rate in prediction. This fusion of CNNs and OLS shows the promise that combining different machine learning techniques can have on the accuracy of biometric recognition.

C. A Novel Approach to Predict Blood Group using Fingerprint Map Reading

Patil et al. [12] introduced a method for predicting blood groups using fingerprint map reading. The method uses the fact that everyone has unique little pattern lines (minutiae) in their fingerprints and these patterns do not change in anyone's life. And through the use of image processing techniques like binary conversion and thinning, the system is supposed to be able to determine a person's blood type without drawing blood, by looking at these patterns, more specifically, the ridge frequency. It uses Multiple Linear Regression with Ordinary Least Squares (OLS) to predict blood type, and is correct 62% of the time.

D. A Novel Approach for ABO Blood Group Prediction using Fingerprint through Optimized Convolutional Neural Network

Vijaykumar et al. [13] describe a novel approach for predicting ABO blood groups from fingerprints using an optimized convolutional neural network (CNN). This method represents a significant breakthrough in biometric analysis, offering a non-invasive and efficient way to determine blood type. If validated, this research could have profound implications for medical diagnostics and transfusion medicine, potentially simplifying the process of identifying a person's blood group and enhancing the safety of procedures involving blood. The proposed CNN design demonstrates notable performance, achieving an accuracy rate of 95.27%.

E. A Cost-Effective Method for Blood Group Detection Using Fingerprints

Siva et al. [14] proposed a new method for determining blood type from fingerprint images. The study focused on enhancing fingerprint-matching efficiency by integrating various feature extraction techniques and employing a Back Propagation Neural Network (BPNN) for classification, achieving an accuracy of 80%. The purpose was to determine an individuals blood type through the comparison of extracted characteristics from fingerprints to a database of known fingerprints.

III. PROPOSED METHODOLOGY

ResNet-50 is well-suited for blood group identificationby analyzing fingerprints due to its deep convolutional architecture, which excels in extracting intricate patterns from images. The network's depth allows it to learn hierarchical features from fingerprint data, aiding in accurate blood group classification. ResNet-50's robustness and effectiveness in image recognition tasks make it an ideal choice for analyzing fingerprint images and predicting blood group types with high accuracy. Leveraging ResNet-50 enables the development of a reliable and efficient system for blood group identification based on fingerprint analysis.The Fig. 1 represents proposed methodology diagram which consists of various phases involved in this study.

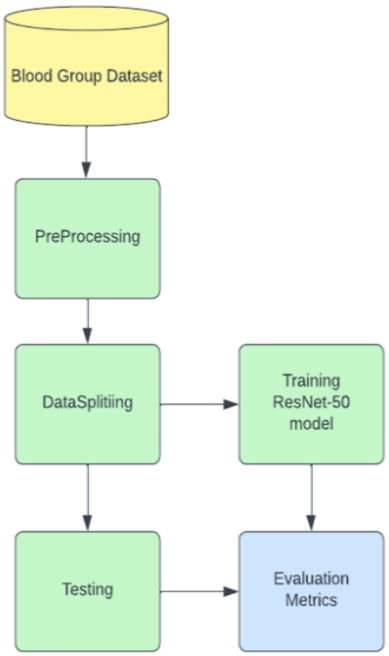

Fig. 1. Methodology diagram

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A. Data Collection

The dataset [15] of fingerprints utilized in this research comprises individual fingerprint images taken by a fingerprint scanner. Each of the fingerprint image is kept in the BMP file format with dimensions with dimensions of 96 by 103 pixels, leading to a file size of roughly 38.8 KB at a resolution of 96 dots per inch (dpi). This dataset encompasses a variety of fingerprint patterns, such as loops, arches, and whorls, from various people. Fig. 2 shows the number of images in each blood group within the fingerprint dataset used in this study.

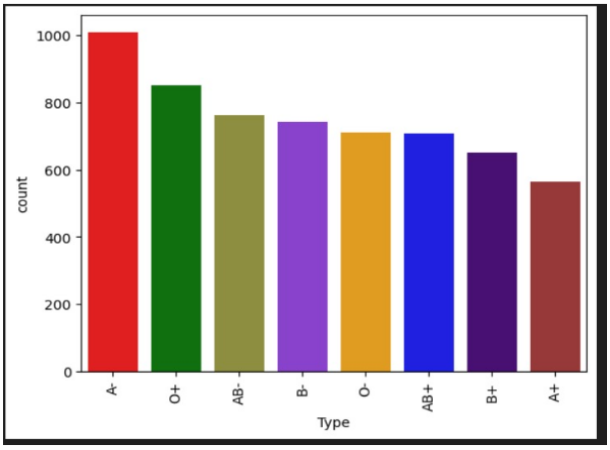

Fig. 2. Dataset

The dataset is structured to include various fingerprint images, and uniquely specific patterns found under fingertips. Each Image is kept by the respective blood group of the individual in different folders, such as A+, A-, B+, B-, O+, O-, AB+ and AB-. Fig. 3 shows images of the dataset used in this study, and labels of each image by blood group of the fingerprints. By compiling a vast collection of images of fingerprints with storage of pictures classified by the person. The dataset on blood groups offers a significant resource for Training and Evaluation of Machine Learning Models focused on Blood type identification by fingerprint analysis.

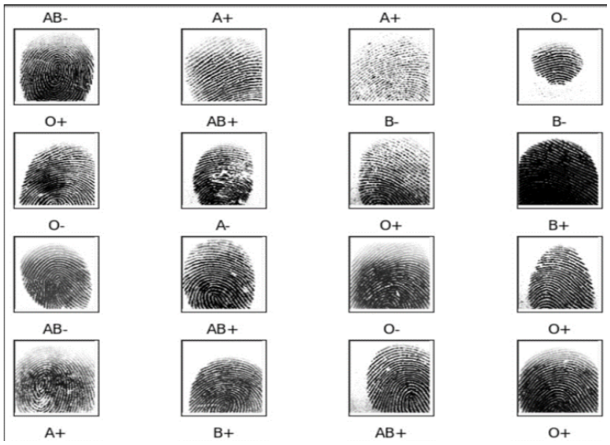

Fig. 3. Data Visualization

B. Data Preprocessing

In this preprocessing, file paths are configured to open up the dataset. The dataset is actually folders, some of which have different blood groups written against them: A+, A-, B+, B-, O+, O-, AB+, AB-. Each blood group folder contains images of fingerprints of people having that type of blood. The images are then transformed into data frames and collected in one form. The processed dataset is divided into training, validation, and testing sets so that the model learns from different examples and can generalize to unseen data through effective training and evaluation of the model.

C. Training and Testing

The training and the testing session involves creating path at file to get the fingerprint database, comparing the file to folders containing different blood groups. Every folder has fingerprints images corresponding to people with specific blood type. Some of the images are then transformed into data frames since this will make the data easy to manage. This is followed by partitioning of data into training (70%), validation and test data to enhance the stability of model learning process. The model utilized here is driven from ResNet-50; and the training data is trained with the pre-processing steps such as resizing the images and normalizing pixel intensity. Subsequently, the performance of the proposed model is evaluated first with the testing image data matrices to measure accuracy as well as the ability to classify the blood group patterns from the fingerprints. By this approach, the approach seeks to accurately and accurately determine the blood group by using deep learning strategies.

IV. Results and Analysis

This approach is based on predicting blood group through fingerprint analysis using ResNet-50: the model development and evaluation process structured. Initially, file paths are made to access the data stored in folders organizing different blood groups, A+, A-, B+, B-, O+, O-, AB+. There are fingerprint images of the various samples corresponding to the individuals with those blood types in each folder.

This data is split into two sets, training set with 75% and testing set with 25%, so that the model trains and tests with accuracy. In fact, the ResNet-50 architecture is indeed very effective in image classification. With the ResNet-50 architecture and with numerous convolutional layers as well as skip connections, it enables the network to learn the complex patterns and features of the images of the fingerprints and therefore predict blood types precisely.

The trained model of ResNet-50 is tested upon testing set to calculate the performance metrics in terms of accuracy and loss. After testing different models multiple times, and gained a maximum accuracy of 88%, which confirms the effectiveness of the ResNet-50 based approach for identification of blood groups from an analysis of fingerprints, thus enriching the construction of valuable

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insights in the field of medical diagnostics and identification purposes. The Fig. 4 shows the output compared to the true data and the predicted image.

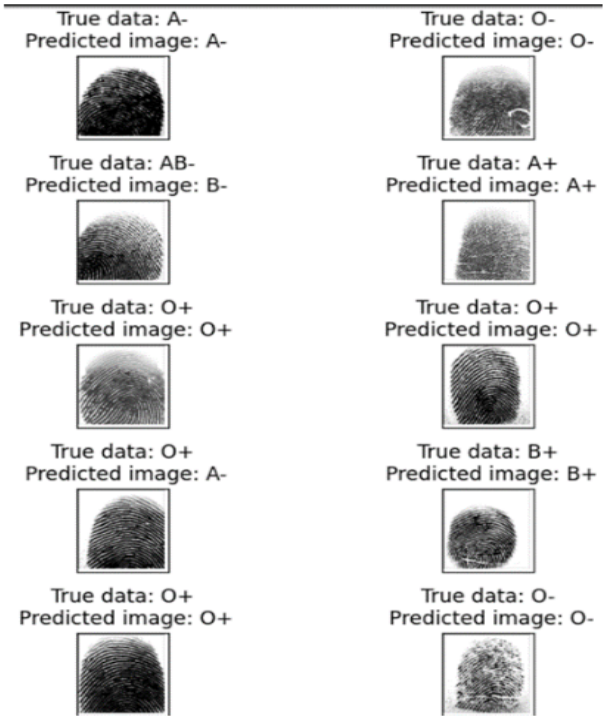

Fig. 4. Actual image v/s Predicted image

This evaluation of the blood group prediction model makes intensive graphical analysis over the learning dynamics and convergence characteristics of it. The training and validation accuracy have been presented to observe its performance epoch-wise. Here, the x-axis represents accuracy levels, and the y-axis, frequency. This visualization, as shown in Fig. 5, helps understand how well this model is good at learning from training data and generalizing to new validation data. Such graphical representation is vital for evaluating the model's development and overall effectiveness.

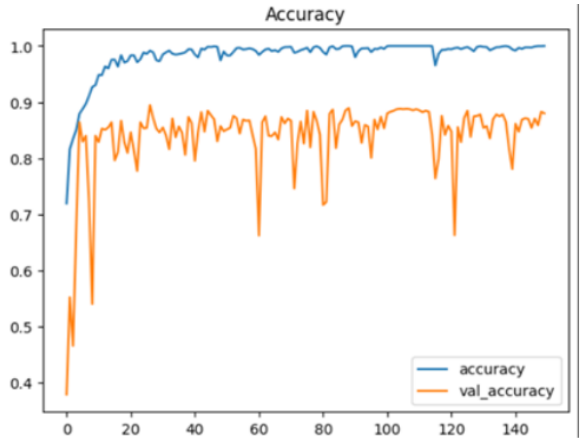

Fig. 5. Training and Validation Accuracy

Additionally, an analysis of the training and validation loss values over epochs is performed, as illustrated in Fig 6. This graph is useful for assessing the model's convergence and performance by monitoring the changes in loss values throughout the training process.

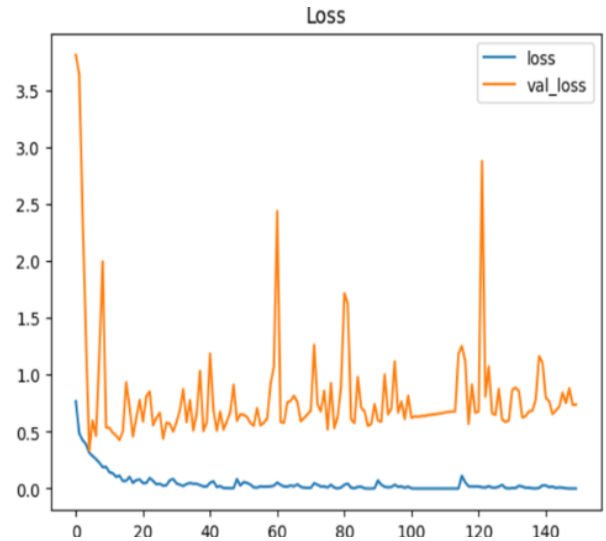

Fig. 6. Training and Validation Loss

The classification report summarizes the model's performance in classification tasks, offering a detailed evaluation. It includes key metrics such as precision, recall, F1-score, and support for each class. Precision reflects true positive predictions made by the classifier. Recall, for a specific class, indicates the proportion of relevant instances correctly identified. The F1-score is the harmonic mean of precision and recall, providing a balanced perspective on performance. Support values show the frequency of each class in the dataset, helping in understanding data distribution. Table I presents the classification report, highlighting these metrics to provide a thorough overview of the model's strengths and weaknesses. This report is crucial for deciding if a model is ready for deployment or needs further optimization, aiding the iterative process of enhancing performance in real-world scenarios.

TABLE I
CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
A+	0.97	0.92	0.95	160
A-	0.86	0.87	0.87	235
AB+	0.91	0.91	0.91	160
AB-	0.84	0.84	0.84	179
B+	0.93	0.87	0.90	164
B-	0.89	0.92	0.90	195
O+	0.84	0.86	0.85	225
O-	0.84	0.85	0.84	182
Accuracy			0.88	1500
Macro Avg	0.89	0.88	0.88	1500
Weighted Avg	0.88	0.88	0.88	1500

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V. CONCLUSION AND FUTURE WORK

Deep learning models, such as ResNet-50, have been shown to hold immense potential for application within biometric identification, specifically within blood group identification from fingerprint images. The implementation of a ResNet-50-based deep learning model has achieved astonishing results with an accuracy level at 88%. This performance underscores the effectiveness of deep learning techniques in automating blood group analysis, offering a more rapid and efficient alternative to traditional methods.

ResNet-50 Architecture: An architecture with advanced feature extraction has been of immense aid in organizing the dataset which includes images from fingerprints categorized by blood groups, namely A+, A-, B+, B-, O+, O-, AB+, AB-. In a structured manner towards the same, it has allowed the model to classify blood types effectively, hence proving the model's viability in the improvement of the accuracy and efficiency of blood-typing procedures.

The future study should address the optimizations of the model architecture and the training procedures. Furthermore, its integration with clinical workflows is essential for more applied usage. By this advancement in blood group identification using a deep learning network based on ResNet-50, much hope is seen for improvement in diagnostic accuracy and efficiency, thus enhancing patient care and services in medicine. The development of such a system represents a significant step forward in leveraging deep learning technologies to streamline and enhance the blood typing process.

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